

Supervised Machine Learning

Naïve Bayes Classifier

CSE4107: Artificial Intelligence

References

- Portland State University, Oregon, United States
- [Analytics Vidhya](#)

Naïve Bayes

- Based on Bayes' theorem
- Statistical classifier
- Supervised learning
- Performance comparable to decision tree and neural network
- High accuracy and speed when applied to large datasets
- **Naïve:** Simplified assumption – 'effect of one attribute is independent of the effect of other attributes'

Naïve Bayes — Bayes Theorem

- Named after Reverend Thomas Bayes
- Based on prior knowledge, describes the probability of an event

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Naïve Bayes — Bayes Theorem

Example

Suppose, a particular test for whether someone has been using cannabis is 90% **sensitive**, meaning the **True Positive Rate** (TPR)=0.90. Therefore it leads to 90% true positive results (correct identification of drug use) for cannabis users.

The test is also 80% specific, meaning **True Negative Rate** (TNR)=0.80. Therefore the test correctly identifies 80% of non-use for non-users, but also generates 20% false positives, or **False Positive Rate** (FPR)=0.20, for non-users.

Assuming 5% **prevalence**, meaning 5% of people use cannabis, what is the probability that a random person who tests positive is really a cannabis user?

- **prevalence** is the proportion of a particular population found to be affected by a medical condition at a specific time.

Naïve Bayes — Bayes Theorem

Example

- Prevalence = 5% = 0.05 = $P(\text{User})$
 - $P(\text{Non-user}) = 95\% = 0.95$
- $P(\text{Positive} \mid \text{User}) = 90\% = 0.9$
- $P(\text{Positive} \mid \text{Non-user}) = 20\% = 0.2$

$$\begin{aligned}
 P(\text{User} \mid \text{Positive}) &= \frac{P(\text{Positive} \mid \text{User})P(\text{User})}{P(\text{Positive})} \\
 &= \frac{P(\text{Positive} \mid \text{User})P(\text{User})}{P(\text{Positive} \mid \text{User})P(\text{User}) + P(\text{Positive} \mid \text{Non-user})P(\text{Non-user})} \\
 &= \frac{0.90 \times 0.05}{0.90 \times 0.05 + 0.20 \times 0.95} = \frac{0.045}{0.045 + 0.19} \approx 19\%
 \end{aligned}$$

Naïve Bayes — Classifier

For Bayes Classifier:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

$$P(A|B) \propto P(B|A)P(A)$$

$$P(A_n \cap A_{n-1} \cap \dots \cap A_1) = P(A_n | A_{n-1} \cap \dots \cap A_1) \cdot P(A_{n-1} \cap A_{n-2} \cap \dots \cap A_1)$$

↑
Chain Rule

For Naïve Bayes Classifier:

- $P(C|X) \propto P(X|C)P(C) = P(X_1, X_2, \dots, X_n | C) \cdot P(C)$
- $P(C|x_1, x_2, x_3, \dots, x_n) = P(C, x_1, x_2, x_3, \dots, x_n) = P(x_1, x_2, x_3, \dots, x_n, C)$
- $= P(x_1|x_2, x_3, \dots, x_n, C_k) \cdot P(x_2, x_3, \dots, x_n, C_k)$
- $= P(x_1|x_2, x_3, \dots, x_n, C_k) \cdot P(x_2|x_3, x_4, \dots, x_n, C_k) \cdot P(x_3, x_4, \dots, x_n, C_k)$
- $= \dots \dots \dots \dots \dots$
- $= P(x_1|x_2, x_3, \dots, x_n, C_k) \cdot P(x_2|x_3, x_4, \dots, x_n, C_k) \cdot \dots \cdot P(x_{n-1}|x_n, C) \cdot P(x_n|C) \cdot P(C)$

Naïve Bayes — Math

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
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D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Naïve Bayes — Math
(Learning Phase)

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Play}=\text{No}) = 5/14$$

We have four variables, we calculate for each

Outlook	Play=Yes	Play=No
<i>Sunny</i>	2/9	3/5
<i>Overcast</i>	4/9	0/5
<i>Rain</i>	3/9	2/5

Temperature	Play=Yes	Play=No
<i>Hot</i>	2/9	2/5
<i>Mild</i>	4/9	2/5
<i>Cool</i>	3/9	1/5

Humidity	Play=Yes	Play=No
<i>High</i>	3/9	4/5
<i>Normal</i>	6/9	1/5

Wind	Play=Yes	Play=No
<i>Strong</i>	3/9	3/5
<i>Weak</i>	6/9	2/5

Naïve Bayes — Math

- **Given** the data as found in last slide:
- **Find** for a new point in space (vector of values) to which group it belongs (classify)

Naïve Bayes — Math (Test Phase)

Test Phase

- Given a new instance of variable values
 $\mathbf{x}' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$
- Given calculated Look up tables

$$\begin{aligned}P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{Yes}) &= 2/9 \\P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{Yes}) &= 3/9 \\P(\text{Humidity}=\text{High} \mid \text{Play}=\text{Yes}) &= 3/9 \\P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{Yes}) &= 3/9 \\P(\text{Play}=\text{Yes}) &= 9/14\end{aligned}$$

$$\begin{aligned}P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{No}) &= 3/5 \\P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{No}) &= 1/5 \\P(\text{Humidity}=\text{High} \mid \text{Play}=\text{No}) &= 4/5 \\P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{No}) &= 3/5 \\P(\text{Play}=\text{No}) &= 5/14\end{aligned}$$

- Use the MAP rule to calculate Yes or No

$$\begin{aligned}P(\text{Yes} \mid \mathbf{x}'): & [P(\text{Sunny} \mid \text{Yes})P(\text{Cool} \mid \text{Yes})P(\text{High} \mid \text{Yes})P(\text{Strong} \mid \text{Yes})]P(\text{Play}=\text{Yes}) = 0.0053 \\P(\text{No} \mid \mathbf{x}'): & [P(\text{Sunny} \mid \text{No})P(\text{Cool} \mid \text{No})P(\text{High} \mid \text{No})P(\text{Strong} \mid \text{No})]P(\text{Play}=\text{No}) = 0.0206\end{aligned}$$

Given the fact $P(\text{Yes} \mid \mathbf{x}') < P(\text{No} \mid \mathbf{x}')$, we label $\mathbf{x}'$ to be “No”.

Naïve Bayes — Continuous Variable

What to do in the case of Continuous Valued Inputs?

- Numberless values for an attribute
- Conditional probability is then modeled with the normal distribution

$$\hat{P}(X_j | C = c_i) = \frac{1}{\sqrt{2\pi}\sigma_{ji}} \exp\left(-\frac{(X_j - \mu_{ji})^2}{2\sigma_{ji}^2}\right)$$

μ_{ji} : mean (average) of attribute values X_j of examples for which $C = c_i$

σ_{ji} : standard deviation of attribute values X_j of examples for which $C = c_i$

- **Learning Phase:** for $\mathbf{X} = (X_1, \dots, X_n)$, $C = c_1, \dots, c_L$
Output: $n \times L$ normal distributions and $P(C = c_i)$ $i = 1, \dots, L$
- **Test Phase:** for $\mathbf{X}' = (X'_1, \dots, X'_n)$
 1. Calculate conditional probabilities with all the normal distributions
 2. Apply the MAP rule to make a decision

Different Naïve Bayes Models

- **Gaussian:** It is used in classification and it assumes that features follow a normal distribution. (*GaussianNB*)
- **Multinomial:** It is used for discrete counts. For example, let's say, we have a text classification problem. Here, we can consider Bernoulli trials which is one step further and instead of “word occurring in the document”, we have “count how often word occurs in the document”, you can think of it as “number of times outcome number x_i is observed over the n trials”. (*MultinomialNB*)
- **Bernoulli:** The binomial model is useful if your feature vectors are binary (i.e. zeros and ones). One application would be text classification with ‘bag of words’ model where the 1s & 0s are “word occurs in the document” and “word does not occur in the document” respectively. (*BernoulliNB*)

Naïve Bayes — Pros and Cons

Pros

- It is easy and fast to predict class of test data set. It also perform well in multi class prediction
- When assumption of independence holds, a Naive Bayes classifier performs better compare to other models like logistic regression and you need less training data.
- It perform well in case of categorical input variables compared to numerical variable(s). For numerical variable, normal distribution is assumed (bell curve, which is a strong assumption).

Naïve Bayes — Pros and Cons

Cons

- If categorical variable has a category (in test data set), which was not observed in training data set, then model will assign a 0 (zero) probability and will be unable to make a prediction. This is often known as **“Zero Frequency”**. To solve this, we can use the **smoothing technique**. One of the simplest smoothing techniques is called **Laplace estimation**.
- Another limitation of Naïve Bayes is the assumption of independent predictors. In real life, it is almost impossible that we get a set of predictors which are completely independent.

Discussion

Naïve Bayes is based on the **independence assumption**

- **Training** is very easy and fast; just requiring considering each attribute in each class separately
- **Test** is straightforward; just looking up tables or calculating conditional probabilities with normal distributions

Naïve Bayes is a popular **generative classifier model**

1. **Performance of naïve Bayes** is **competitive** to most of state-of-the-art classifiers even in presence of violating independence assumption
2. It has many successful applications, e.g., **spam** mail filtering
3. A good candidate of a base learner in **ensemble learning**
4. Apart from classification, naïve Bayes can do more...